



# **Aras Innovator 33**

## **Configurable Job Scheduling**

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# 1 Overview

Define and manage one-time and recurring jobs in Aras innovator without installing additional modules.

Aras Configurable Job Scheduling service allows you to define and manage jobs that run once or on a recurring schedule. Use this runtime-agnostic service to schedule action endpoints in your application or long-running processes using Conversion tasks or external services. Use REST APIs to schedule jobs, including long-running jobs asynchronously, and create multiple schedule formats for simple and complex recurring schedules. Manage jobs and tasks and manage schedules with a web-based user interface.

## 1.1 Configurable Job Scheduling Key Item Types

The Configurable Job Scheduling allows administrators to define and manage jobs that run once or on a recurring schedule from a centralized Aras Innovator web-based user interface.

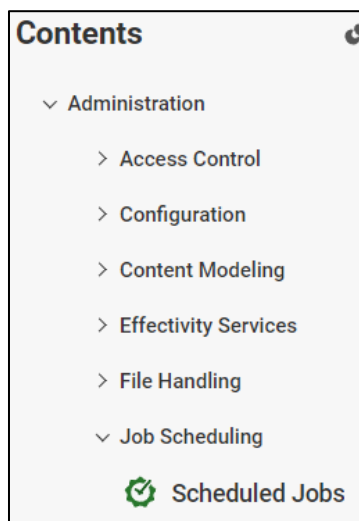
The basic data model consists in two main ItemTypes:

1. **Scheduled Job:** Is created by Aras Innovator administrators and defines both the Job Execution Schedule and the Job to be executed.
2. **Job:** Is created by the scheduler at any recurrency of the Job Execution Schedule

## 1.2 Permissions

Configurable Job Scheduling Job is intended to be a tool for Aras Innovator Administrators or other Innovator system functions and applications.

Scheduled Job Items are added by default to the TOC under **Administration → Job Scheduling**



### 1.2.1 Users and Roles

Each Job has a User property which grants its own permissions for executing its Server Method. This would potentially allow any user to be assigned to Job or Scheduled Job. To avoid a user impersonating another user, Aras Innovator introduces special system users and roles.

### 1.2.2 Roles

- **World** – Any User in Aras Innovator. No permission to create Jobs and Scheduled Jobs. Implemented by core World identity.
- **Administrators** – Aras innovator Administrators. Has permission to create, modify and delete Scheduled Jobs. Has permissions to create Jobs special endpoints, get and delete them. Implemented by core Administrators identity.
- **Job Manager** – Configurable Job Scheduling system identity. Can Create and Modify Jobs, get Jobs Schedule and execute Scheduled Jobs

### 1.2.3 Users

- **Innovator User** – Any Innovator user. Can be assigned to Job and Scheduled Job. Should be able to manually run only Jobs that are assigned. Cannot create Jobs or Scheduled Jobs if not part of Administrators identity.
- **Agent User** – A special system user without password, represents an Agent Service which sends requests to Innovator Server. Allowed to get Scheduled jobs and execute them. This user is assigned by default to Job Manager role.

## 2 Required Configuration for Scheduled Jobs

### 2.1 Configure Agent Service Pulling Interval Configuration

Aras Innovator Agent Service is extended with a background service to execute Jobs. See *Section 4 – Scheduled Jobs Processing*.

Agent Service has one instance of `JobsBackgorundService` per each Innovator Database. To configure which databases should be processed. `JobsProcessingCycles` section should be added to `appsettings.json` config file of Agent Service:

```
"JobsProcessingCycles": [
{
  "DB": "{DB1}",
  "Interval": 2
},
{
  "DB": "{DB2}",
  "Interval": 20
}
```

Each background service requires two settings:

- **DB** – defines a database to fetch scheduled jobs from
- **Interval** – defines a pulling interval in seconds to fetch Scheduled Jobs. This configuration is optional. If not set the default value is set to 5.

## 3 Configuring Job Scheduling

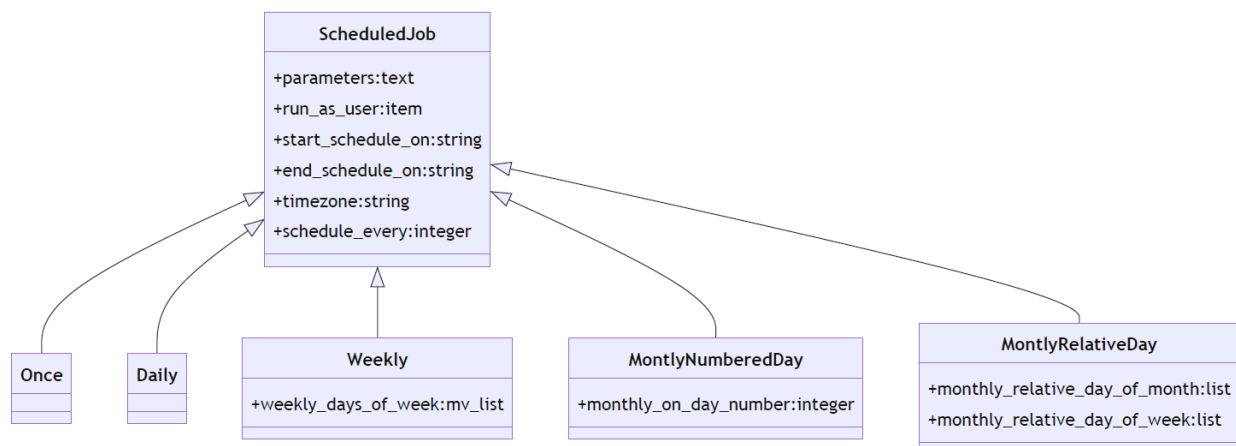
### 3.1 Scheduled Job

**Scheduled Job** Is created by Aras Innovator administrators and defines both the Job Execution Schedule and the Job to be executed.

It supports the following scheduling options:

- **Once** – *Job* is created at a given time only once.
- **Daily** – *Jobs* are created at a given time each day until expiration of the schedule.
- **Weekly** – *Jobs* are created at a given time of selected days of the week, each week until expiration of the schedule.
- **Monthly, Numbered Day** – *Jobs* are created at a given time of selected days of the week, each week until expiration of the schedule.
- **Monthly, Relative Day** – *Jobs* are created at a given time of selected days of the month, each month until expiration of the schedule.

These options are implemented using ItemType Classification. Each of them adds its own specific properties and has its own View defined.





## 3.2 Configure Scheduled Job

The Scheduled Job Item implements some common properties and extends for different Schedule types.

### 3.2.1 Scheduled Job shared properties

The common properties are:

- **Name**
- **Run As User**
- **Parameters**
- **Start Schedule On**
- **Time Zone**

#### 3.2.1.1 Run As User property

Defines an Innovator User whose behalf Job method will be executed. Configurable Job Scheduling impersonates this user during Job's execution. It allows to set Permissions context scoped to the defined User. Run As User property is a required property.

#### 3.2.1.2 Parameters property

Contains **json** dictionary which is used to execute *Jobs*. Implemented as `Text` property.

Parameters is a json dictionary

```
{
  "method_name": "method to execute name",
  "method_param_1": "parameter to send to a method in method_name"
}
```

**method\_name** is a required parameter to define the Server Method Name to be executed. The json can contain any number of parameters to be sent to the defined method when it is called.

**Note:** The Text approach for Parameters is used to add flexibility to the data model and support not only Server Methods.

#### 3.2.1.3 Start Schedule On property

Defines date and time used to calculate first Job creation time. Implemented as a `string` property in format `dd/MM/yyyy HH:mm:ss`. Start Schedule On property is a required property.

#### 3.2.1.4 Time Zone property

Defines the time zone applicable to any date and time defined in the Scheduled Job Item.

Implemented as a populated `list`.

**Note:** Since client and server can be in different Time Zones. The explicit definition of the selected Time Zone for the dates properties allows consistency. All dates input as strings are converted to UTC format using the value set in the Time Zone property.

### 3.2.2 Once Schedule

Once is convenient way of preparing a Job execution to happen in the future at a given time.

The screenshot shows the configuration interface for 'Scheduled Job 1'. The 'Schedule' dropdown is set to 'Once'. The 'Start Schedule On' field is empty, and the 'Time Zone' is set to 'W. Europe Standard Time (UTC+01:00)'. The 'Run As User' field is also empty. The 'Parameters' field is a large text area.

### 3.2.3 Daily Schedule

Daily Schedule is a recurrent schedule set to repeat every given number of days at the **Start Schedule Time**.

The screenshot shows the configuration interface for 'Scheduled Job 1'. The 'Schedule' dropdown is set to 'Daily'. The 'Start Schedule On' field is empty, and the 'Time Zone' is set to 'W. Europe Standard Time (UTC+01:00)'. The 'Run As User' field is also empty. The 'Parameters' field is a large text area. The 'End Schedule On' field is empty.

Given its recurring nature the **End Schedule On** property is required to define a limit to close the cycle.

### 3.2.4 Weekly Schedule

Weekly Schedule is a recurrent schedule set to repeat every given number of weeks at the **Start Schedule Time** on a specific day, or set of days, of the week.

The screenshot shows the 'Scheduled Job 1' configuration window. At the top, there are buttons for 'Save', 'Done', 'Delete', and several icons. Below this is a 'Scheduled Job' tab. The form contains several fields: 'Name' (yellow), 'Run As User' (yellow), 'Start Schedule On' (yellow), 'End Schedule On' (grey), 'Time Zone' (dropdown menu showing 'W. Europe Standard Time (UTC+01:00)'), and 'Schedule' (dropdown menu showing 'Weekly'). There is also a 'Parameters' text area on the left. On the right, there are additional settings: 'Every' (dropdown menu showing 'Weeks') and 'On' (dropdown menu showing days of the week: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday).

Given its recurring nature the **End Schedule On** property is required to define a limit to close the cycle.

### 3.2.5 Monthly, Numbered Day Schedule

Monthly, Numbered Day Schedule is a recurrent schedule set to repeat every given number of months at the **Start Schedule Time** on a specific day number of the month.

The screenshot shows the 'Scheduled Job 1' configuration window. At the top, there are buttons for 'Save', 'Done', 'Delete', and several icons. Below this is a 'Scheduled Job' tab. The form contains several fields: 'Name' (yellow), 'Run As User' (yellow), 'Start Schedule On' (yellow), 'End Schedule On' (grey), 'Time Zone' (dropdown menu showing 'W. Europe Standard Time (UTC+01:00)'), and 'Schedule' (dropdown menu showing 'Monthly, Numbered Day'). There is also a 'Parameters' text area on the left. On the right, there are additional settings: 'Every' (dropdown menu showing 'Months') and 'On Day Number' (text input field).

Given its recurring nature the **End Schedule On** property is required to define a limit to close the cycle.

### 3.2.6 Monthly, Relative Day Schedule

Monthly, Relative Day Schedule is a recurrent schedule set to repeat every given number of months at the **Start Schedule Time** on a day defined as First, Second Third, Fourth of Last, day or days of the week.

Given its recurring nature the **End Schedule On** property is required to define a limit to close the cycle.

## 3.3 Scheduled Job Lifecycle

A default Lifecycle is assigned to each Scheduled Job Item.

### 3.3.1 Scheduled Job Life Cycle Map

The lifecycle controls the states of the Scheduled Job controlling editing permission.



Only Job Manager role is assigned to the transitions from one state to the other.

Possible states are:

- **Inactive** – The Scheduled job is configured to start in the future. In this state all the properties of the Scheduled Job can be edited by Aras Innovator Administrators.
- **Activated** – The Scheduled Job is running. It means the *Start Schedule On* date is passed and at least first Job execution already happened. In this state only *Run As User*, *Parameters* and *End Schedule On* properties can be edited by Aras Innovator Administrators.
- **Deactivated** – The *End Schedule On* date is passed and the scheduler completed its cycle. In this state the Item cannot enter in Edit mode and no property can be updated anymore, even by Aras Innovator Administrators.

## 4 Scheduled Jobs Processing

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Scheduled Jobs are just templates of Jobs with a schedule to define when the Job must be instantiated autonomously.

Once a Job is created by `CreateJob` Server Method, it is saved in the database and published to the queue for processing. To manage asynchronous execution, Job execution is implemented with an in-memory queue.

A special background service implemented as an extension of the innovator **Agent Service** is set to listen to the queue and process Jobs from it. This Service has a limited number of parallel threads to limit server overload.

Every Job is then processed by a *Worker Thread* as a self-request to Aras Innovator Server to execute the Server Method configured in the Job.

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**Warning** Since *Job Execution Queue* is implemented as Innovator Server in-memory queue, it is possible that Jobs might remain in Awaiting status if Innovator server crashes, is restarted due to IIS reset or autoscaler event.

**Warning** *Worker Thread* sends requests to Innovator Server. Failed Jobs can happen in case of network issues or timeouts.

**Warning** Jobs can remain In Progress state even if the actual job execution completed successfully. This can happen in case of critical failure of Innovator Server, IIS reset or lost connection to database when the job is completing execution.

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## 4.1 Job properties

Job is a unit of work to be executed asynchronously from its creation moment. Job is implemented via ItemType Job.

| Property       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID             | Unique identifier of Job                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Parameters     | Contains a Json dictionary passed by the configured Scheduled Job see <i>Section 3.2.1.2 - Parameters property</i>                                                                                                                                                                                                                                                                                                                                                         |
| UserID         | Run As User 's User, as configured in the Scheduled Job<br>see <i>Section 3.2.1.1 - Run As User property</i>                                                                                                                                                                                                                                                                                                                                                               |
| Status         | Represent current status of Job. Implemented as <code>list</code> property.<br>Following States are implemented: <ul style="list-style-type: none"> <li>• <b>Awaiting</b> – Job is created, not taken for execution yet.</li> <li>• <b>In Progress</b> – Job is being executed.</li> <li>• <b>Failed</b> – Job method returned error result or an unhandled exception was thrown during method call.</li> <li>• <b>Success</b> – Job was successfully executed.</li> </ul> |
| Result Message | Contains the text with Job execution result if server method returns result, otherwise contains details of the thrown exception.                                                                                                                                                                                                                                                                                                                                           |
| Scheduled For  | Time of scheduled execution. Implemented as date property. It is stored as UTC time in the database and usually transformed to Client time zone when is fetched from Innovator Server.                                                                                                                                                                                                                                                                                     |
| Started On     | Time of actual execution start and corresponding to the transition to In Progress status. Implemented as date property. It is stored as UTC time in the database and usually transformed to Client time zone when is fetched from Innovator Server.                                                                                                                                                                                                                        |
| Finished On    | Time when Job execution completed and corresponding to the transition to either Success or Failed status. Implemented as date property. It is stored as UTC time in the database and usually transformed to Client time zone when is fetched from Innovator Server.                                                                                                                                                                                                        |

## 5 Configurable Job Scheduling vs Scheduler Service

Configurable Job Scheduling

| Modules                                            | Configurable Job Scheduling                 | Scheduler Service               |
|----------------------------------------------------|---------------------------------------------|---------------------------------|
| <b><u>Infrastructure</u></b>                       |                                             |                                 |
| <b>Required Components</b>                         | Aras Innovator Server<br>Aras Agent Service | Aras Innovator Server           |
| <b>Requires separate installation</b>              | NO                                          | YES                             |
| <b>Cross Platform</b>                              | YES                                         | NO                              |
| <b><u>Administration</u></b>                       |                                             |                                 |
| <b>Configuration Management</b>                    | Aras Innovator Web Client                   | Configuration file on Scheduler |
| <b>Monitoring in Aras Innovator</b>                | YES                                         | NO                              |
| <b><u>Scheduler</u></b>                            |                                             |                                 |
| <b>One Time Execution</b>                          | YES                                         | YES                             |
| <b>Daily, Weekly, Weekly<br/>Relative, Monthly</b> | YES                                         | YES                             |
| <b>Subintervals (seconds,<br/>minutes, hours)</b>  | NO                                          | YES                             |